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Employing Google Trends and Deep Learning in Forecasting Financial Market Turbulence

Anastasios Petropoulos, Vasileios Siakoulis, Evangelos Stavroulakis, Panagiotis Lazaris, and Nikolaos Vlachogiannakis

Bank of Greece

ABSTRACT

In this paper we apply text mining methodologies on a set of 10,000 Central Bank speeches to construct a financial dictionary, based on which we use Google Trends indices to measure people's interest in financial news. Particularly, we investigate the relationship between these indices and financial market turbulence leveraging on Deep Learning techniques, which are benchmarked against a variety of Machine Learning algorithms and traditional statistical techniques. Our main finding is that Google queries convey information able to predict future market turbulence in a short time period (one month), and that Deep Learning algorithms clearly outperform over benchmark techniques. Google Trends can provide useful input in the creation of crisis Early Warning Systems, as social data are more responsive compared to official financial indicators, which are usually available with a lag of several weeks or months. Thus, such an Early Warning System (EWS) that is continuously updated with current social data can be a valuable tool for policymakers, as it can immediately identify signs of whether a crisis is imminent or not.

KEYWORDS

Forecasting; Google Trends; XGBOOST; Deep Neural Networks; Text mining

Introduction – motivation

In today's digital world, the internet is the primary source of information. Many people around the globe, especially from modern economies, use various web search engines to hunt for almost any kind of information. Google is the most popular web search engine, handling more than five billion searches on a daily basis. Google gathers sheer volume of data about the search terms, which are widely available through Google Trends (GT), in a specific website that analyses the frequency and popularity of queries in Google. This analysis may reveal economic, social, and health trends. Specifically, Google Trends has been found to correlate with unemployment levels and flu outbreaks, and most importantly provide the information faster compared to official statistics and traditional reporting. On top of that, Natural Language Processing (NLP) has been gaining ground as an additional financial analysis tool in the recent years, mainly driven by the evolution of machine learning techniques and big data. Following this trend, we depart from the conventional macroeconomic analysis, and use as input the historical time series of Google search terms instead of macro variables with the purpose to

examine per se the hit rate of a model solely based on Google searches data.

We use Google Trends indices to measure the people's interest in financial news, so that the relationship between these indices and market turbulence to be investigated. We therefore assume that Google searches include information able to predict future market turbulence. Our work provides further evidence about the importance of the information conveyed by social data, as social data is more responsive compared to official financial indicators, which are usually available with a lag of several weeks or months. Thus, a responsive Early Warning System (EWS) continuously updated with current social data, is a valuable tool for policymakers, as it can identify immediate signs as to whether a crisis is impending. Hence, policy makers may unveil vulnerabilities of the economy sooner, and take precautionary actions to effectively diminish the risks that can trigger a crisis.

In this study, we create a time series of Google searches for a set of specific terms, and use them to forecast turbulence events. The time series are based on indices provided by Google Trends, on a scale from 0 to 100, representing the monthly relative

search volume, covering the period from 2004 to 2018. The terms are certain words, bigrams, and trigrams, selected from a specific dictionary including 3,749 terms. This dictionary was created by means of NLP sentiment analysis to Central Bank speeches ranging from 2008 to 2018. Our dependent variable is the increased market turbulence, which is constructed based on three of the key stock market indices of the world (i.e. DAX, SPX, NKY). In particular, a market turbulence event is defined if two out of the three indices fall by more than 5% within a month. The forecast horizon is set to one month.

Our contribution to the literature about Google Trends and their connection to the recorded performance of financial markets is summarized in the following points. First, our results provide evidence on the increased significance of the information conveyed by global Google search queries. Secondly, we create a financial term dictionary based purely on text mining techniques on a set of 10,000 Central Bank speeches delivered by senior central bank and banking institutions officials. Third, from a technical perspective we build a group of state-of-the-art statistical machine and deep learning techniques, including Machine Boosting algorithms and Deep Neural Networks, in the classification problem of future market crash detection. Finally, this is a novel solution to the problem of forecasting economic crises, by developing a complementary and more responsive Early Warning System based on Google searches information that precedes official statistics and macroeconomic indicators in the short run.

The remainder of the paper is structured as follows. In the second section, we review the literature on Google Trends analysis and forecasting. The third section describes the data related processes used (text mining, variable space reduction), and the fourth section analyses the methodology employed. The fifth section presents the empirical results of our analysis using Google Trends indices. Results focus on the predictive ability of the machine learning approach employed, and performance is measured both using in sample and out sample data. The sixth section concludes the paper and provides recommendations on further research.

Literature review

The ever-increasing availability of data via the internet has recently turned researchers into more frequently making use of it into their empirical work. Internet data is abundant, and typically not structured in such

a way that can be easily incorporated in quantitative analysis. However, a notable exception is Google Trends, which provides standardized figures in a structured way, based on the number of searches for various keywords. This is the reason why Google Trends has become popular in recent research analysis, with already a wide range of applications across most scientific fields, including health, medicine, IT, social sciences, as well as economy and financial markets (Jun, Yoo, and Choi 2018). In addition, as Jun, Yoo, and Choi (2018) notice, the current research has shifted the focus from the ex post analysis of trends, to the forecasting of future changes.

Below we present an overview of studies that make use of Google Trends data in the modeling of macroeconomic or financial variables. To begin with, various researchers have investigated whether the use of internet data can improve the prediction of foreign exchange rate movements. Bulut (2018) used weekly Google Trend data in order to forecast the exchange rate moves. He compared his approach to structural macroeconomic models and concluded that Google Trends search query data outperform the structural models in predicting the actual direction of monthly changes in nominal exchange rates, both during the period of 2004–2009 and during the stressed global economic environment of 2008–2009. Masuda and Takeda (2019) conducted an analysis focusing in USD-JPY currency pair. They constructed sentiment indices for Japan and United States in order to capture the sentiment prevailing in those markets, which were then used in structural and autoregressive models as additional explanatory variables. These models exhibited higher predictive power, by decreasing the mean squared prediction errors. In the same vein, Wilcoxson, Follett, and Severe (2020) explored the relationship of United States Dollars with 10 other currencies during the time period of 2004–2018 and found evidences that Google Trend query can assist in predicting the aforementioned relationship. However, the authors noticed that certain scalability adjustments are required on the coefficients of Google Trend data.

Dergiades, Milas, and Panagiotidis (2015) investigated the relationship of credit spread for certain European countries with the related discussions in social media (Twitter, Facebook and Google blogs), as well as the search queries related to the Greek debt crisis. They concluded that the abovementioned internet information is an important addition for the credit spreads of the Greek and Irish Governments, even if other financial variables are already taken into account. Whereas, internet information was of lower

significance for the credit spreads of Portugal, Italy and Spain. Along the same lines, González-Fernández and Carmen (2018) analyzed how internet search activity, measured through Google data, influenced the sovereign bonds yields of five core and five peripheral European countries. They discovered that Google data are positively correlated with the yield of Greek sovereign bond.

Other researchers focus on examining the relationship between internet information and stock market volatility. Vlastakis and Markellos (2012) suggested that the request of information of the 30 largest stocks traded on NYSE and NASDAQ, approximated by the Google search volume, is a driver of future volatility. At the same time, Aouadi, Arouri, and Roubaud (2018) posit that information demand, as approximated by the search volumes in Google Trends, is positively correlated with stock market liquidity. Kostopoulos, Steffen, and Charline (2020) examined how Google search volumes related to the economic concerns of households (FEARS index), affected the trading behavior of individuals with internet access. They found that individual investors disinvest of risky assets when the FEARS index is high, with the less sophisticated of them appearing to be more susceptible to the sentiment index. Kristoufek (2013) proposed a diversification portfolio strategy based on the internet popularity of stocks, which outperformed other conventional portfolio strategies. Vozlyublennaya (2014) highlighted the relationship between investor attention, as expressed through Google search queries on security indexes, and the asset prices dynamics. It was noticed that, when there is a short-term change in index returns, it is following by an increase in investor attention, and in tandem, a shock on returns leads to a long-term change in attention. Moreover, a strong correlation between volatility and attention was observed. Given that volatility and investor attention show strong correlation on short horizons, Hamid and Heiden (2015) incorporated the search engine data in their models. This way, they managed to improve the out-of-sample volatility forecasting, compared to conventional time series models. Similar results are also found by Yung and Nafar (2017), who posit that Real Estate Investment Trusts (REITs) exhibiting high retail investor attention, as measured by search volumes in Google Trends, generate higher returns relative to REITs with no retail investor attention.

Other researchers attempted to construct some type of “early warning indicators” based on internet data. Preis, Moat, and Stanley (2013) set up an investment strategy in Dow Jones Industrial Average, based on

the relative change in search volumes of a specific set of terms. The performance of these so-called Google Trend strategies indicates that, among others, they may be used as “early warning signs” of stock market moves, hence, increasing the chances for investors to profit from their trades. Heiberger (2015) developed an investment strategy that was profiting in times of financial turmoil by using firm-specific volumes of Google search queries as a “bad news” indicator. The success of this strategy indicates that this information could be used as an early warning system from financial system risk. Hu et al. (2018) assessed the performance of an improved sine cosine algorithm relative to a series of other models. All the models were assessed both including and excluding Google Trends information. They found that their model is superior in predicting the opening direction of S&P and Dow Jones Industrial Average in both cases. On top of that, they note that when their model includes Google Trends information, its hit ratio reaches almost 90%.

Naccarato et al. (2018) incorporated internet query information into econometrics models that were used to predict Italian youth unemployment rate. The additional internet information conveyed extra explanatory power, as its inclusion decreased the forecasting error of the models. Similar results are presented by Simionescu, Streimikiene, and Strielkowski (2020), who forecasted the unemployment rate in UK, using a panel data model with internet and official data. The aforementioned estimation outperformed the forecasts that were solely based on official indicators. Yilmazkuday (2020) investigated the relationship between Google search queries of “coronavirus” and “unemployment,” using daily data from the United States. He found that interest in the word “unemployment” is mostly explained by that of “coronavirus” in March 2020. That is, there are some interlinkages between the COVID related measures taken by governments around the world to limit the loss of human lives and the impact on global economies. D’Amuri and Marcucci (2017) examined the use of a Google job-search index as a leading determinant for predicting the monthly US unemployment rate. In the out-of-sample forecasting they found that models including Google data outperformed most of the other models, while at the same time, their predictive performance increased with the horizon of the forecast. At the other end of the spectrum, the results of Nagaoa et al. (2019) didn’t indicate a de facto improvement of the US unemployment rate prediction when internet Google query information was fed in the widely used autoregressive models.

Typically, most of the economic indicators are becoming available with a time lag. For instance, business cycle turning points are generally released with a lag of at least 5 months. However, it is critical for the policy makers to access such information the soonest possible. To this end, Chen et al. (2015) found the 2007 peak turning point timely and accurately, using Markov Switching Model and Google queries information. Dietzel (2016) achieved to produce similar results in terms of accuracy and timeliness, when forecasted the turning points in the housing market using sentiment indicator based on internet query data. Kohns and Bhattacharjee (2019) managed to improve the performance of nowcast US real GDP models, by including search terms in the models specification. Whereas, Niesert et al. (2020) used Google search data to predict a series of macroeconomic variables, including unemployment, Consumer Price Index and consumer confidence, for a series of developed countries. They focus on the out of sample performance of their models, to find evidence that only in predicting unemployment the explanatory power is retained. Others posit that Google Trends data are also important determinants in forecasting oil prices (Lu et al. 2020) and oil consumption (Yu et al. 2019).

Considering that all the above-mentioned papers highlight the additional explanatory power that can be derived from variables constructed by using internet data, we make use of it, so as to forecast financial market turbulence. To the best of our knowledge, we are not aware of any other study that investigates the aforementioned questions as such, or implements various modeling approaches (ranging from simple logistic regression, to Machine Learning, and to Deep Learning) in order to identify the approach that maximizes the hit rate in predicting a financial crisis.

Data collection, processing and variable selection

In order to identify the relevant terms that will be used in the development of the historical time series of Google searches, we have initially created the corresponding dictionary. The dictionary of the relevant financial terms is based on text analysis of a corpus of central bank speeches ranging from 2008 to 2019. This period includes a series of financial market turbulence events, including the 2008 financial crisis when Lehman Brothers collapsed. This period was selected due to the fact that central banks significantly altered their policy and became more transparent in their communication to the public. These speeches are

collected from the official BIS website. Our dataset is comprised of more than 10,000 central bank speeches delivered by senior central bank and institutions officials at various events. Their content is rather broad, covering financial and economic matters, issues related to the respective central bank entity (e.g. insurance schemes, organization issues), and topics that are delivered at universities to inspire students.

The central bank speeches collected in our dataset have a global jurisdiction coverage (see Appendix: Table A.1), and as is presented in Figure 1 below, these speeches are almost uniformly distributed across the whole period of the analysis.

Text analysis of the Central Bank speeches

We tokenized the set of documents focusing on monograms, bigrams and trigrams as they refer to phrases with a clear sentiment indication in the context of a speech. Noise removal is crucial in this process as special characters, symbols, alpha-numeric characters, numeric characters, urls, emails, and, most importantly, stop words like: “and,” “or,” “the” should be omitted as they do not add any significance to the relevant speech. In addition, tokens are also filtered for names, like the speaker’s surname, and words included in the header and footer of the document, which tend to occur repeatedly without any sentiment meaning.

In the corpus of speeches used in the analysis, there were a lot of speeches regarding non - financial related topics, like speeches in the graduation of a University or speeches about some specific ceremony events (anniversary of currencies for example) in some countries we omitted irrelevant documents based a Latent Dirichlet Allocation (LDA) which is a popular method for fitting a topic model. The output of the LDA fitted model¹ is a probability of each word to belong to each of the recognized topics. Based on the implementation of the algorithm two topics are recognized (i.e. “economic and monetary policy” & “financials and central banking”). From a relevance perspective since both topics are relevant for our analysis, the higher the probability to belong in topic A or B the higher the relevance. On the other hand, documents/speeches that are less condensed in terms of belonging to a specific topic (A or B) have low probability to be relevant for sentiment analysis. In all we excluded the documents with low probability to belong to those two topics recognized.

Furthermore, not only documents, but also terms that exhibit low probability to belong in either topic

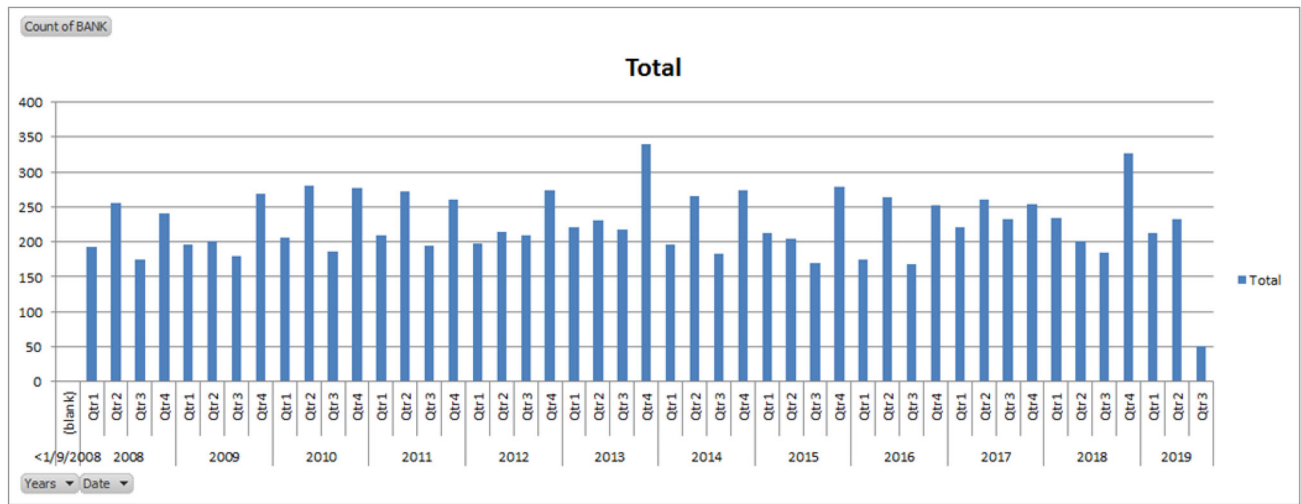


Figure 1. Distribution of speeches across the analysis period. Logistic Regression (Logit), Conditional Inference Tree (CIT), RF (Random Forest), SVM (Support Vector Machine), XGBOOST (Extreme Gradient Boosting), DNN (Deep Neural Network), BSNN (Bayesian Deep Neural Network).

where excluded (threshold applied 0.015) since the analysis attempts to find terms relevant to finance or economics for projecting the probability of a stock market turbulence event.

In order to measure the importance of each term in our corpus of documents we measure their frequency of appearance. For example, for a specific trigram, e.g. “financial market turbulence,” we count the number of appearances across all speeches. The more frequent the term is, the higher its relevance to the whole corpus content, so that a higher score for relevance is assigned to a speech that contains a highly frequent trigram. We employ the tf-idf statistic in order to measure the importance of a term (monograms, bigrams and trigrams) across the whole collection of central bank speeches. The formula for the estimation of tf-idf² statistic is given below:

$$w_{i,j} = tf_{i,j} \times \log \left(\frac{N}{df_i} \right)$$

$tf_{i,j}$ = number of occurrences of i in j df_i = number of documents containing i N = total number of documents

The TF-IDF allocates high probability to terms that are important within each document in the corpus. By applying a threshold of 0.01 on the TF-IDF metric we select the most significant terms for the context of each document. The highest scoring words of a document are the most relevant to that document, and therefore they can be considered keywords for that document. Furthermore, by grouping all together the terms we can construct a set of keywords for the whole corpus. Based on that, less importance is assigned in terms that are not globally important base

on the tf-idf statistic and higher importance is assigned to terms with a higher appearance frequency making it more probable to be relevant for inclusion in the analysis.

The TF_IDF and the LDA criteria where applied together to filter a set of term out of a large amount of words which are both relevant to the topics recognized by the LDA and are also importance in the context of each document meaning using TF-IDF. These decreased significantly the terms which initially where approximately 3749 terms excluding the stop words like: and, or, the etc. We also include the month to month percentage changes of the search frequency indexes for the selected terms, leading to a total of 7498 variables in total.

Variable reduction

Fitting a machine learning model to a huge number of independent variables (relative to the size of the dataset) is doomed to suffer from the so-called curse of dimensionality problem, whereby the fitted classifier may seem to yield very good performance in the training dataset, but it turns out to yield very poorly, in the test data.

Thus, to ensure a good performance outcome for our model, we need to implement a robust independent variable (feature) selection stage so as to limit the number of used features to the absolutely necessary. Apart from increasing the generalization capabilities of the fitted models, such a reduction is also important for increasing the computational efficiency of the explored machine learning algorithms.

Our dependent variable is the increased market turbulence which is identified based on the performance of three of the key stock market indices of the world (i.e. DAX, SPX, NKY). A turbulence event is realized if two out of the three aforementioned indices fall by more than 5% within a month. The forecast horizon is set equal to one month.

Our variable selection strategy consists in estimating the Pearson's Correlation coefficient since the two variables are numeric. Specifically, we consider the dependent variable which is binary as numeric taking the value of 1 in a crisis event and 0 otherwise. We follow a univariate analysis, in the first step, by selecting variables which exhibit some level of positive or negative correlation with the dependent variable in order to decrease the noise in the data and increase afterwards the efficiency of a multivariate selection second step implemented on the basis of a Boruta algorithm. Decreasing the dimensionality through the univariate analysis is deemed appropriate in order to exclude non correlated variables and improve the robustness of the Boruta algorithm to select important variables for a multivariate analysis setup.

In particular we run a correlation analysis of the 7498 variables with the target variable, dropping out all the terms whose frequency of appearance exhibited a lower than 10% correlation (in absolute value) with the target variable. This step reduced the total number of terms to 2367. In the next step we employ the Boruta algorithm to independently assign importance to the available features.

The Boruta algorithm³ is based on a postulated Random Forest model. Boruta infers the importance of each independent variable (feature) in the obtained predictive outcomes by creating shuffled copies of all features (shadow features) and then fitting a Random Forest on the extended dataset. In each step the algorithm checks whether a real feature has a higher importance than the best of its shadow features based on Z-score, and constantly removes features that are deemed unimportant. The algorithm stops when all features are classified as important or are rejected as noise. In this way, all features relevant to dependent variable are selected based on error minimization for the fitted Random Forest models, in each iterative step of the algorithm. From the Boruta variable selection process, 30 terms out of 2367 candidates were selected for the development analysis, alleviating dimensionality issues (Appendix – Table A.2).

Model development

In our effort to examine whether the trends in Google search of specific terms provide valuable input in the

forecasting of financial turbulence, we assess the final selected search terms in a series of statistical methods. Namely, we include in our analysis: Logistic Regression (henceforth LogR), Random Forests (henceforth RF) (Breiman 2001), Support Vector Machines (henceforth SVMs) (Vapnik 1998), Conditional Inference Trees (henceforth CIF) (Strasser and Weber 1999), extreme gradient boosting (henceforth XGboost) (Friedman 2001), and two Deep Learning techniques, namely Deep Neural Networks (DNN) and Bayesian Neural Networks (BSNN).

We implemented logistic regression by using the GLM function that performs optimization through Iteratively Reweighted Least Squares. For SVM's we evaluate soft-margin SVM classifiers using linear, radial basis function (RBF), polynomial, and sigmoid kernels, and retain the model configuration yielding optimal performance. The candidate values of these hyperparameters are selected based on a grid-search algorithm (Vapnik 1998). For all other methodologies we performed an extensive cross-validation procedure, based on 75% – 25% segmentation rule of the training set to select a series of entailed hyper parameters. The area under the curve (AUROC) metric was used for model selection in the context of cross-validation due to the binary nature of the dependent variable.

Especially for Deep Neural Networks we employ Dropout DNNs with ReLU and Local Winner-Takes-All (LWTA) activations to train and deploy feed forward deep neural networks. We assess different structures up to five hidden layers deep comprising from various numbers of neurons. The difference of Bayesian Deep Neural Networks lies in the imposition of a prior Normal distribution over network weights, seeking to infer their posterior distribution given the data. Since the marginal likelihood is intractable for the considered architectures, from the existing Bayesian methods, we rely on approximate inference and specifically on Variational Inference. Since the true posterior of the model cannot be computed in Variational Inference, we introduce an auxiliary variational posterior distribution of a family of distributions and then try and find the optimal member of the considered family to match the true posterior. This matching is achieved through the minimization of the Kullback-Leibler (KL) divergence between the true and the introduced variational posterior.

Model validation

In order to assess the robustness of our approach we perform a thorough validation procedure. We assess

Table 1. Results of the metrics for the out of sample performance of all candidate models.

	Logit	CIT	RF	SVM	XGBOOST	BSNN	DNN
AUROC	0,82	0,85	0,86	0,91	0,86	0,94	0,97
KS	0,51	0,71	0,76	0,79	0,78	0,88	0,95
BA	0,75	0,83	0,79	0,86	0,71	0,89	0,89
WBA	0,71	0,83	0,81	0,84	0,61	0,93	0,88
Y	0,50	0,66	0,59	0,71	0,43	0,82	0,77
DP	1,25	1,76	1,51	1,98	1,40	2,75	2,26
LM	0,40	0,20	0,22	0,19	0,54	0,66	0,14
LP	3,90	4,88	3,42	6,83	6,83	7,21	8,62
G	0,74	0,83	0,79	0,86	0,68	0,91	0,89

AUROC: Area Under the Curve, KS: Kolmogorov-Smirnov test, BA: Balanced Accuracy, WBA: Weighted Balanced Accuracy, Y: Youden Index, DP: Discriminant Power, LM: Negative Likelihood Ratio, LP: Positive Likelihood Ratio, G: Geometric Mean.

the out-of-time performance by splitting our sample based on the 75 – 25 rule, i.e. the train sample covers the period from 2004 to 2015 and the test sample covers the period from 2016 to 2019. More precisely, we report the performance results obtained from the experimental evaluation of our method, in terms of in-sample fit and out-of-sample performance. Note that, in order to properly calibrate the hyper parameters of the machine learning algorithms in the current study, the in-sample dataset is further split randomly in train and validation set using the 75 – 25 rule. The validation dataset is used to find the best set of hyper parameters of the models and select the best candidate model for performing the out-of-sample evaluation.

We employ a series of metrics that are broadly used for quantitatively estimating the discriminatory power of each scoring model, such as the Area under the ROC curve metric, as well as the Kolmogorov Smirnov (KS) statistic. The KS statistic is measuring the difference in the distribution of scores which are the output of the models trained/validated in our study. The scores provided by most of the models in the analysis are a continuous variables. Specifically, the KS statistics in the analysis is defined as the Kolmogorov–Smirnov statistic for the difference between the cumulative distribution of the crisis events and the distribution of non-crisis events in our dataset based on the score of each model.

In addition, considering that financial crisis events are not a common phenomenon, there is a predominance of non-stressed market periods in our subsamples. That is, our dataset is strongly imbalanced, in the sense that it is not evenly split between crises and non-crisis. Imbalanced data learning is one of the most challenging problems in data mining due to their skewed class distribution which may provide misleading classification accuracy. We therefore used a crisis probability cutoff point according to which we separate the predicted crisis and non-crisis periods.

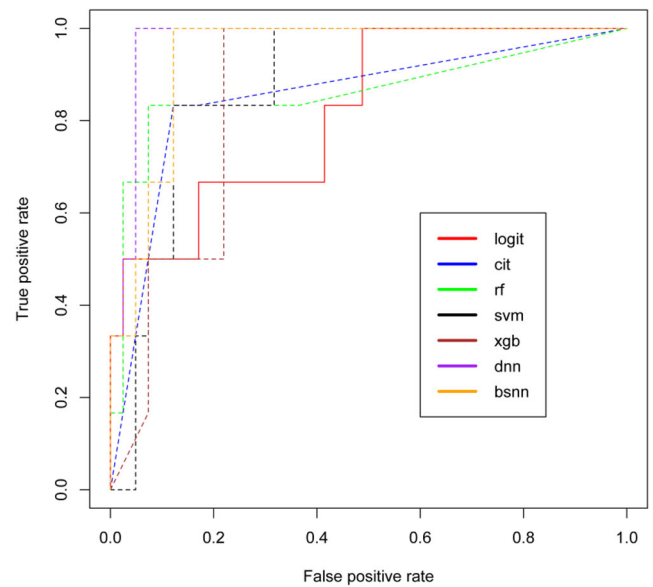


Figure 2. ROC curve for forecasting a financial crisis event. Logistic Regression (Logit), Conditional Inference Tree (CIT), RF (Random Forest), SVM (Support Vector Machine), XGB (Extreme Gradient Boosting), DNN (Deep Neural Network), BSNN (Bayesian Deep Neural Network).

After thorough examination of different values for this parameter, and based on the performance of the classification in the short in-sample dataset used for model development, we set the cut off criterion based on a cost profile function where we penalize a false positive 5 times against a false negative. Based on that we calculate a series of metrics proposed by Bekkar, Khelouane, and Taklit (2013) described analytically in the Appendix Part C.

Based on the above results in Table 1 we note that Deep Learning methodologies outperform all benchmarks models, along almost all measures. Our final assessment is based on the AUROC and KS metrics since they are classification metrics broadly used for all rating systems, and do not depend on the cutoff point used.

Furthermore, we present in Figure 3 the ROC curves corresponding to the methodologies analyzed. This curve is created by plotting the true positive rate against the false positive rate at various threshold settings. As such, they illustrate the obtained tradeoffs between sensitivity and specificity, as any increase in sensitivity will be accompanied by a decrease in specificity. The closer the curve follows the left-hand border and then the top border of the ROC space, the more accurate the modeling approach. The corresponding ROC curve of extreme gradient boosting (XGBoost) is higher over all the considered competitors supporting the high degree of efficacy and generalization capacity of the proposed employed machine learning system.

It is evident from Figure 2, that Deep Learning classification lines are closer to the top left corner of the graph. Also, all Machine Learning algorithms such as SVM, XGBoost and Random Forest clearly outperform traditional classification methods such as Logistic Regression. Even though in terms of AUROC Deep Learning models are better, all other techniques employed allow us to gain useful insight relevant to

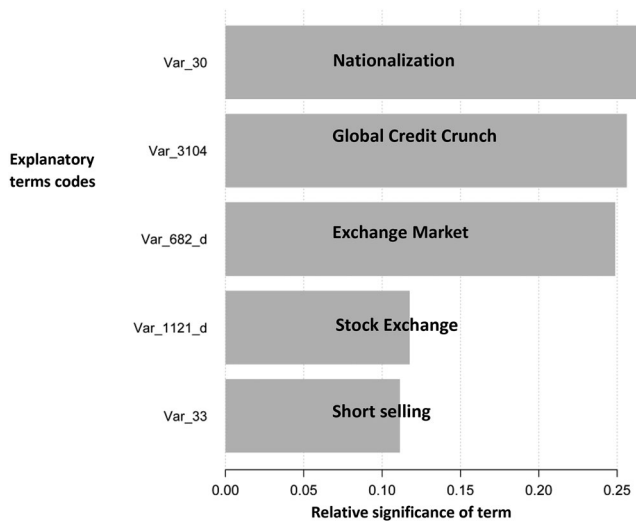


Figure 3. XGBoost variable importance plot top 5 variables. The x-axis describes the percentage contribution of the predictor in the “real” model (Var_30: “Nationalization,” Var_3104: “Global Credit Crunch,” Var_682: “Exchange Market,” Var_1121_d: “Stock Exchange,” Var_33: “Short Selling”). “_d” sub refers to percentage change in the search frequency index across months.

financial crisis forecast. In particular Boosting (XGBOOST) and Bagging (Random Forest) methodologies have the relative advantage that they are not “black boxes” regarding the factors affecting the final result, since they provide a module for calculating variable importance measures through a reshuffling process. In other words, after predicting with the benchmark model, the reshuffling technique produces hundreds of predictions for each variable in the model while randomizing that variable. If the variable being randomized affects the model’s benchmark score, then it is ranked as a high importance variable whereas if on the other hand the loss is minimal then the variable is ranked as low importance. Therefore, we run the variable importance algorithm and we show in Figure 3 the ranked list of the most important variables. Increasing Google search of terms such as “Global Credit Crunch,” “Short Selling” and “Nationalization” provides strong signal for an impending financial turbulence.

Similar insights are gained from Figure 4, where we show the selected Conditional Inference Tree, indicating that a large number in the searches of the term “Global Credit Crunch” (>37 in the Google Trend index) leads to a large probability of financial crisis. Also, even if the search frequency of the term “Global Credit Crunch” is less than 37 then increasing search frequency of the term “Bear Market” provides a weaker but nonetheless significant signal of financial crisis. Finally, a search frequency above >12 of the term “Nationalization” provides a weaker crisis signal.

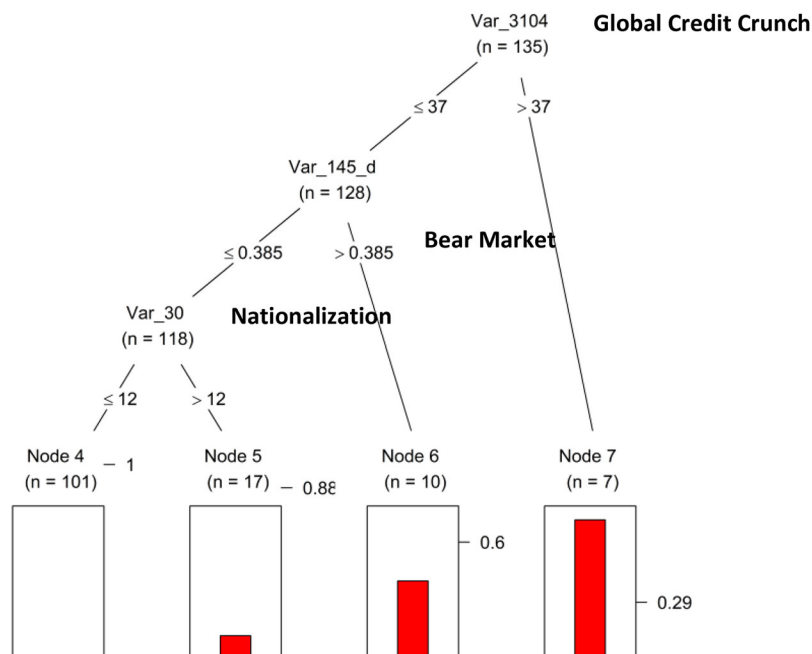


Figure 4. Conditional inference tree for the prediction of financial crisis. (Var_3104: “Global credit crunch,” Var_145_d: “Bear Market,” Var_30: “Nationalization”). _d sub refers to percentage change in the search frequency across months.

Table 2. Logistic regression coefficients for financial crisis period classification problem _d Sub refers to percentage change in the search frequency across months.

Coefficients:	Estimate	Std.Error	z-value	Pr(> z)
Nationalization	0,234	0,137	1,702	0,089
bank governors	−0,539	0,161	−3,356	0,001 ***
global credit crunch	0,062	0,045	1,375	0,169
euro zone_d	5,730	2,698	2,124	0,034 *
deposit guarantee scheme_d	2,263	1,012	2,236	0,025 *
short term gains_d	−8,691	4,249	−2,046	0,041 *
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				

The above mentioned terms, are in agreement also with the finally selected logistic regression coefficients (Table 2), which include the terms “Nationalization” and “Global Credit Crunch” with a sign indicating the positive relation between their appearance frequency and crisis probability along with the reverse impact of positive terms search such as “short term gains.”

Finally, in order to further illustrate the performance of the trained models we present the classification tables in the out-of-sample dataset, using an optimal sensitivity vs specificity weighted cut off for classifying crisis relative to non-crisis periods. A False Alarm is defined as the case where the model predicts a crisis and the prediction is not verified. Hit Rate is defined as the cases where the model predicts a crisis and the prediction is verified. Based on Table 3, it is clear that in terms of hit rate, Deep Learning methods reach 100% hit rate with comparatively low False Alarm rate. All other ML models, reach a 5 to 6 crisis accuracy (83%), but the SVM model has the lowest False Alarm rate compared to competitor models, justifying the better benchmarking metrics (12%). All the other models provide a False Alarm metric in the range of 20% to 22%. It is therefore evident that Deep Learning algorithms can achieve better out-of-sample predictive accuracy when compared to other benchmark methods in the framework of imbalanced test datasets (comparatively small number of crisis periods).

Conclusions and future work

In this paper we apply Text Mining methodologies on a set of 10,000 Central Bank speeches to construct a financial dictionary, based on which we use Google Trends indices to measure the people’s interest in financial news. We investigate the relationship between these indices and financial market turbulence leveraging on Deep Learning techniques, which are benchmarked against a variety of Machine Learning algorithms and traditional statistical techniques.

Table 3. Classification tables of all candidate models.

Logit	pred			
TRUE	No-crisis	Crisis	Signal	Rate
No-crisis	33	8	False Alarm	20%
Crisis	2	4	Hit Rate	67%
CIT	pred			
TRUE	No-crisis	Crisis	Signal	Rate
No-crisis	34	7	False Alarm	17%
Crisis	1	5	Hit Rate	83%
RF	pred			
TRUE	No-crisis	Crisis	Signal	Rate
No-crisis	33	8	False Alarm	20%
Crisis	1	5	Hit Rate	83%
SVM	pred			
TRUE	No-crisis	Crisis	Signal	Rate
No-crisis	36	5	False Alarm	12%
Crisis	1	5	Hit Rate	83%
XGBOOST	pred			
TRUE	No-crisis	Crisis	Signal	Rate
No-crisis	32	9	False Alarm	22%
Crisis	1	5	Hit Rate	83%
BSNN	pred			
TRUE	No-crisis	Crisis	Signal	Rate
No-crisis	33	8	False Alarm	20%
Crisis	0	6	Hit Rate	100%
DNN	pred			
TRUE	No-crisis	Crisis	Signal	Rate
No-crisis	36	5	False Alarm	12%
Crisis	0	6	Hit Rate	100%

Our main finding is that Google queries convey information able to predict future market turbulence in a short time period (one month), and that Deep Learning algorithms clearly outperform benchmark techniques. At the same time, seeing that macroeconomic data are published with a lag, so that market developments usually precede the signal that a macro model might provide, we provide a novel solution to the problem of forecasting economic crises. Leveraging on public source information (e.g. Google queries) we provide the basis for developing a more responsive Early Warning System, which could provide valuable input for the short-term market dynamics to supervisors and practitioners. It has to be noted that the detection of appropriate search term is critical in building such models, since failing to detect proper search terms will affect the accuracy rate of an Early Warning System. That said, our methodology builds on a robust term selection process, which isolates the terms based on their relevance in central bank speeches with the use of text mining techniques.

As a proposal for future work, one could further boost the term selection process by employing it to online data so that the terms are updated constantly based on their frequency of appearance in the internet news. This selection process could lead to a set of indices that are regularly fed in the network, which in turn will provide impending crisis signals. Finally, the inclusion of macro and market data along with their interaction with Google Trends, in the creation of an

Early Warning System, is a topic that could also be proven really interesting for market practitioners.

Notes

1. The algorithm is implemented using the package “topicmodels” in R.
2. We employ the tidytext package in R in order to apply tf-idf analysis.
3. We employ the Boruta Package, provided by the R programming language.

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Appendix

Table A.1. Distribution of speeches by Central Bank.

Central Bank	Speeches in the sample
Bank for International Settlements	1
Bank Indonesia	49
Bank of Albania	198
Bank of Algeria	4
Bank of Argentina	17
Bank of Aruba	1
Bank of Austria	33
Bank of Bahamas	9
Bank of Bahrain	36
Bank of Barbados	70
Bank of Belgium	13
Bank of Belize	1
Bank of Bolivia	1
Bank of Bosnia and Herzegovina	4
Bank of Botswana	33
Bank of Brazil	8
Bank of Cambodia	1
Bank of Canada	296
Bank of Chile	92
Bank of China	31
Bank of Colombia	1
Bank of Curacao and Saint Maarten	13
Bank of Cyprus	3
Bank of Denmark	57
Bank of England	429
Bank of Estonia	13
Bank of Finland	107
Bank of France	208
Bank of Ghana	31
Bank of Greece	106
Bank of Guatemala	1
Bank of Guyana	2
Bank of Hungary	9
Bank of Iceland	54
Bank of India	9
Bank of Ireland	201
Bank of Israel	62
Bank of Italy	210
Bank of Jamaica	15
Bank of Japan	417
Bank of Jordan	1
Bank of Kenya	145
Bank of Korea	59
Bank of Kosovo	6
Bank of Kuwait	2
Bank of Latvia	3
Bank of Lithuania	12
Bank of Luxembourg	26
Bank of Macedonia	65
Bank of Malaysia	329
Bank of Malta	27
Bank of Mauritius	101
Bank of Mexico	77
Bank of Mozambique	1
Bank of Namibia	20
Bank of Nepal	14
Bank of Netherlands	84

(Continued)

Table A.1. Continued.

Central Bank	Speeches in the sample
Bank of Nigeria	17
Bank of Norway	130
Bank of Pakistan	67
Bank of Papua New Guinea	37
Bank of Philippines	280
Bank of Portugal	50
Bank of Romania	51
Bank of Russia	16
Bank of Samoa	5
Bank of Serbia	58
Bank of Seychelles	8
Bank of Sierra Leone	4
Bank of Slovakia	3
Bank of Slovenia	4
Bank of Solomon Islands	13
Bank of Spain	166
Bank of Sri Lanka	30
Bank of Tanzania	1
Bank of Thailand	141
Bank of Trinidad and Tobago	81
Bank of Turkey	62
Bank of Uganda	128
Bank of Ukraine	7
Bank of United Arab Emirates	5
Bank of Uruguay	1
Bank of Zambia	104
Bulgarian National Bank	22
CPMI	3
Croatian National Bank	6
Czech National Bank	32
Deutsche Bundesbank	468
Eastern Caribbean Central Bank	6
European Central Bank	1,586
Federal Reserve Bank of Atlanta	2
Federal Reserve Bank of Boston	5
Federal Reserve Bank of Chicago	8
Federal Reserve Bank of Dallas	19
Federal Reserve Bank of Kansas City	17
Federal Reserve Bank of Minneapolis	23
Federal Reserve Bank of New York	277
Federal Reserve Bank of Philadelphia	32
Federal Reserve Bank of Richmond	2
Federal Reserve Bank of San Francisco	1
Federal Reserve System	702
Hong Kong Monetary Authority	105
Maldives Monetary Authority	4
Monetary Authority of Macao	22
Monetary Authority of Singapore	168
Reserve Bank of Australia	323
Reserve Bank of Fiji	99
Reserve Bank of India	584
Reserve Bank of Malawi	21
Reserve Bank of New Zealand	85
Reserve Bank of Vanuatu	1
Saudi Arabian Monetary Agency	17
South African Reserve Bank	219
Sveriges Riskbank	191
Swiss National Bank	201
Total	10,538

Table A.2. Final terms obtained through the variable selection process.

Term code	Search term
Var_30	Nationalization
Var_33	Short selling
Var_149	Capitulation
Var_231	Credit crunch
Var_281	Stock market
Var_308	Financial meltdown
Var_810	Market turmoil
Var_1175	Bank governors
Var_1276	Deposit guarantee
Var_3104	Global credit crunch.x
Var_3273	Global credit crunch.y
Var_3523	Financial crisis hit
Var_2_d	vix_d
Var_30_d	nationalization_d
Var_33_d	short selling_d
Var_145_d	Bear market_d
Var_281_d	stock market_d
Var_451_d	exchange rate_d
Var_454_d	global financial_d
Var_538_d	exchange rates_d
Var_682_d	exchange market_d
Var_1121_d	stock exchange_d
Var_1133_d	euro zone_d
Var_1150_d	credit default_d
Var_1490_d	exchange_d
Var_1506_d	finance_d
Var_1546_d	currency_d
Var_2242_d	deposit guarantee scheme_d
Var_2305_d	deposit insurance scheme_d
Var_3192_d	short term gains_d

The "d" sub refers to percentage change in the search frequency across months.

Part CI Evaluation Measures

Defining Sensitivity and Specificity as follows:

$$Sensitivity = \frac{TP}{TP + FN}, \quad Specificity = \frac{TN}{TN + FP}$$

where:

TP = True Positive, TN = True Negative, FN = False Negative and FP = False Positive.

We can define the following metrics (Bekkar, Khelouane, and Taklit 2013)

Index	Index Name	Formula
G-mean	Geometric mean	$\sqrt{(sensitivity * specificity)}$
LM	Negative likelihood ratio	$(1 - sensitivity) / specificity$
LP	Positive likelihood ratio	$sensitivity / (1 - specificity)$
DP	Discriminant power	$\sqrt{3/\pi} [\log(sensitivity/(1-sensitivity)) + \log(specificity/(1-specificity))]]$
BA	Balanced accuracy	$1/2 (sensitivity + specificity)$
WBA	Weighted balanced accuracy	$0.75 * sensitivity + 0.25 * specificity$
γ	Youden's index	$sensitivity - (1 - specificity)$